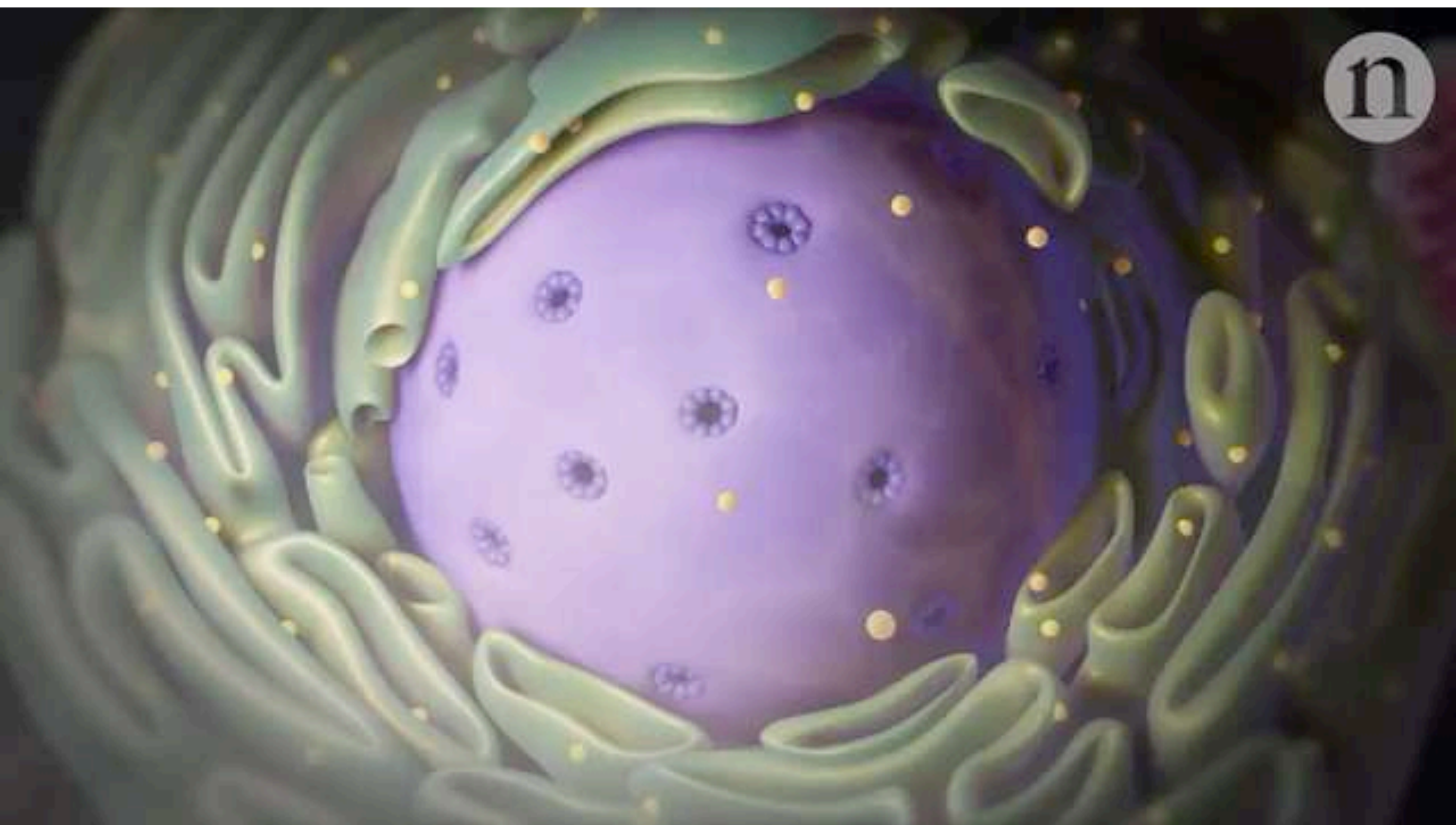
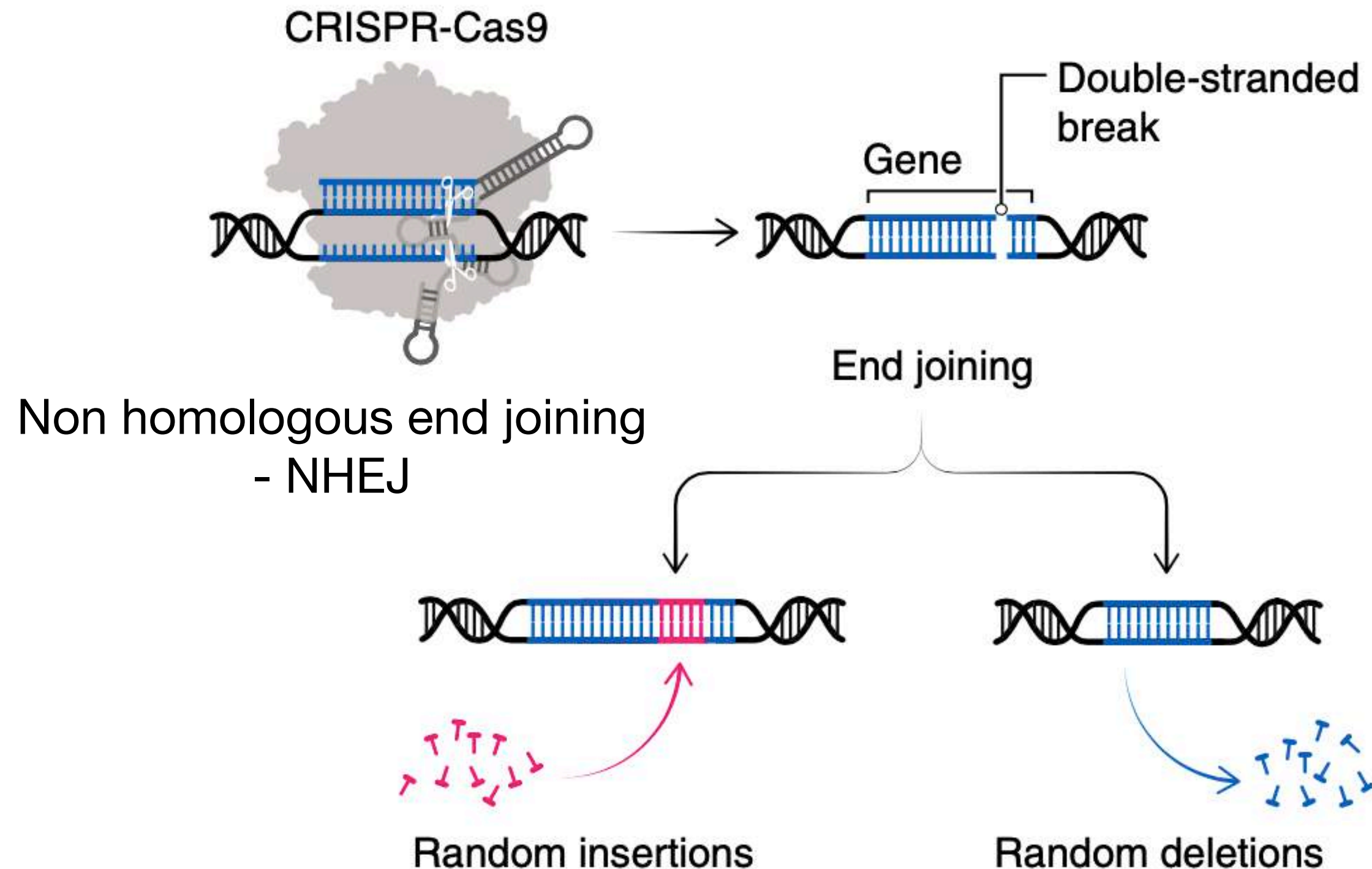




<https://www.youtube.com/watch?v=4YKFw2KZA5o>

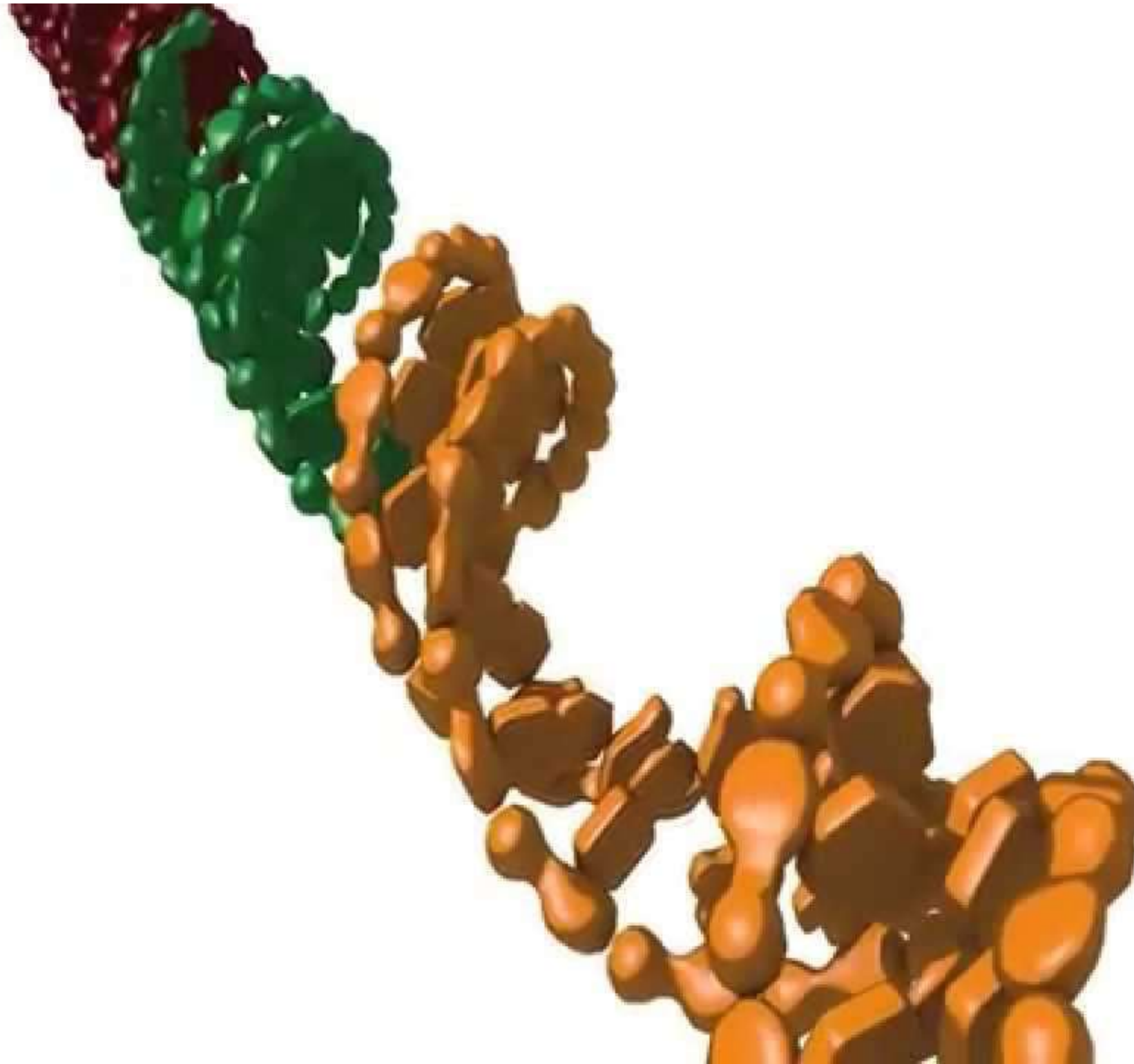
Gene editing by CRISPR - disrupt the gene

DNA breaks caused by cas9 are repaired without external information



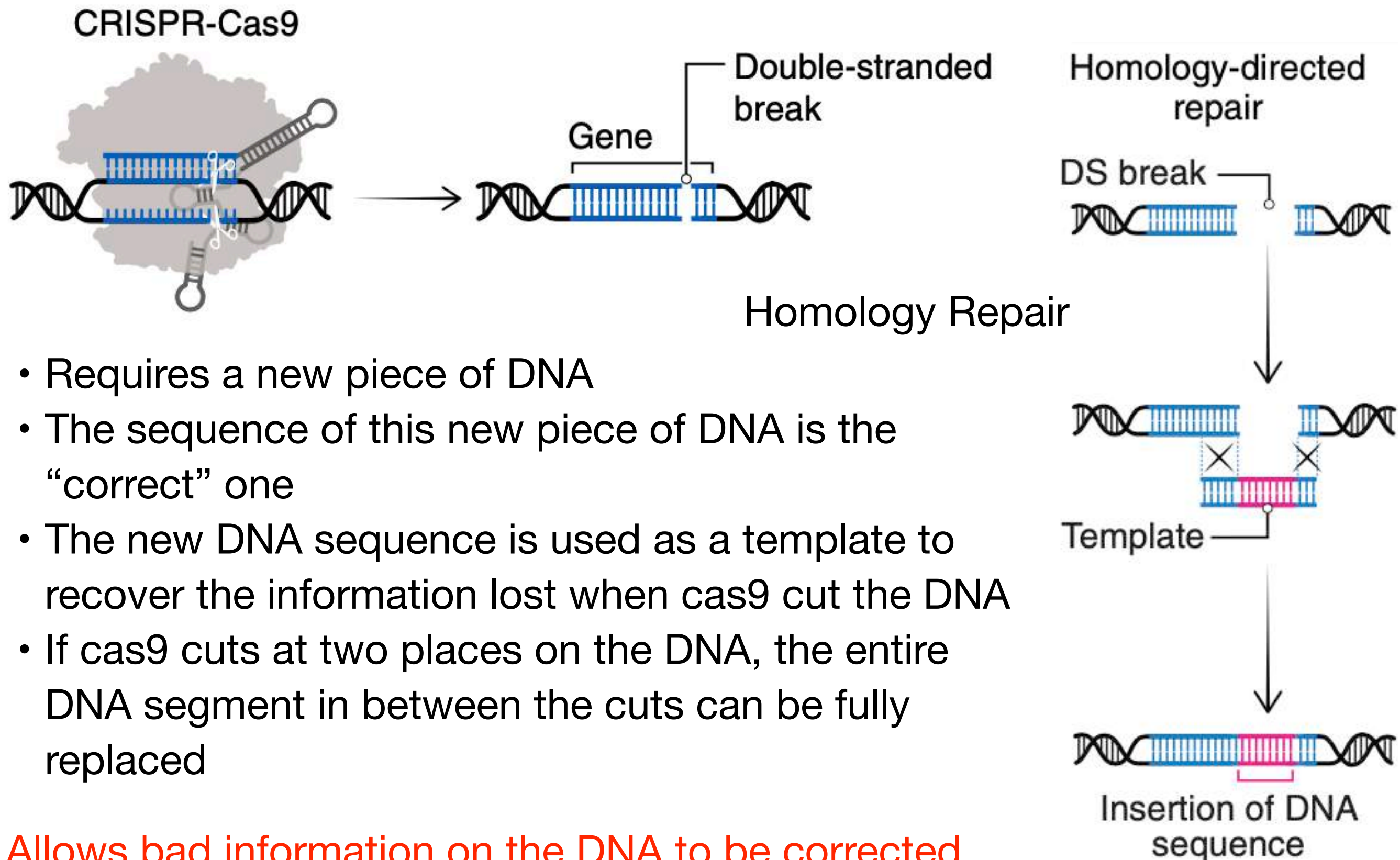
Few bases can get added or removed at the cut site - alters the original information in a disruptive manner

Non homologous end joining - NHEJ

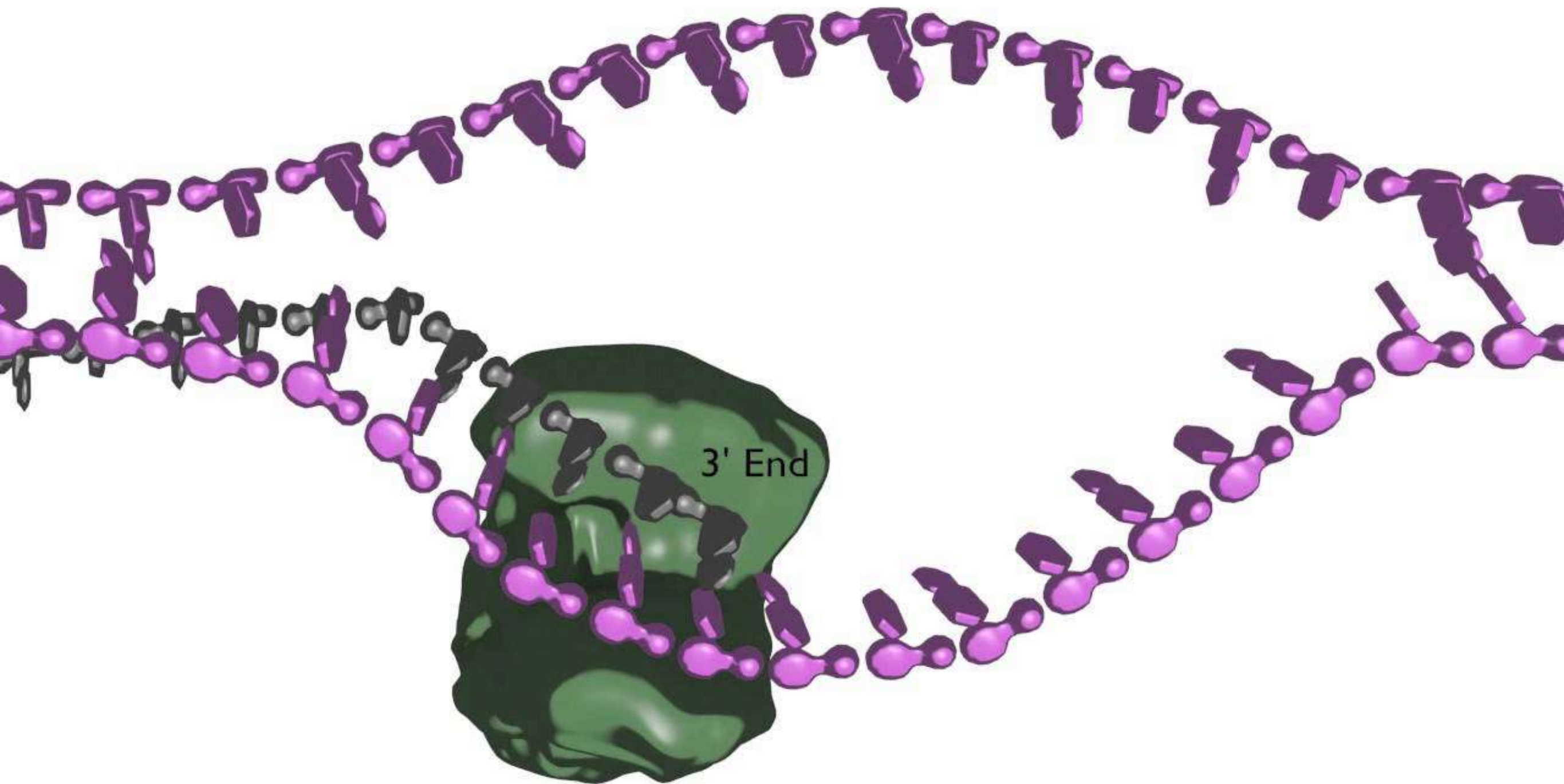


Gene editing by CRISPR - repair the gene

DNA breaks caused by cas9 are repaired with the aid of external information



Homology Directed Repair



<http://www.ensembl.org/index.html>

Human

HBB gene

Understand how to navigate the database

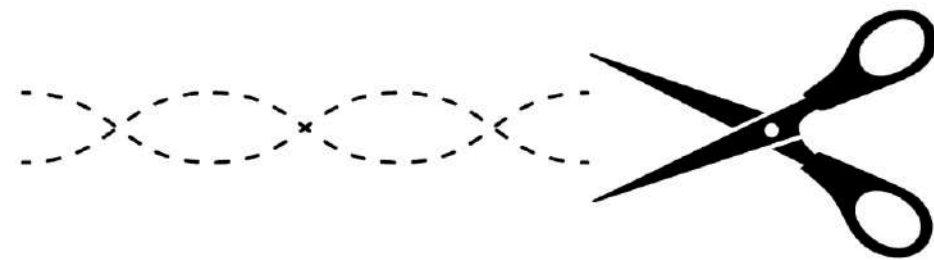
NAME YOUR DATABASE

If you had a database where anyone could check and design guideRNAs for their genes of interest, what would you name it?

<https://chopchop.cbu.uib.no/#>

[Home](#)[Instructions](#)[Scoring](#)[About](#)[Updates](#)[Submissions](#)[Contact](#)[FAQ](#)

CHOPCHOP



Target

RefSeq/ENSEMBL/gene name or genomic coordinates.

In

[Add](#) new species.

Using

Change default PAM and guide length in Options.

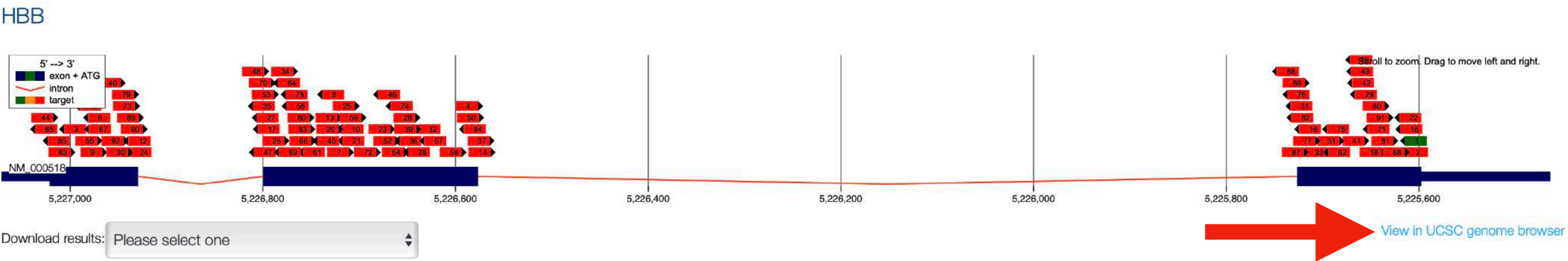
For

- ✓ knock-out
- knock-down
- knock-in
- activation
- repression
- nanopore enrichment

[Paste Target](#)[Options](#)[Reset Options](#)[Find Target Sites!](#)

Guide RNAs to target HBB gene in *H sapiens*

Read instructions page and understand what the various columns are



Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	MM0	MM1	MM2	MM3	Efficiency
1	GCGAGCTTAGTGATACTTGTGGG	chr11:5225592	+	45	0	0	0	0	0	55.74
2	AGCGAGCTTAGTGATACTTGTGG	chr11:5225591	+	45	0	0	0	0	3	62.63
3	GTAACGGCAGACTTCTCCTCAGG	chr11:5226984	+	55	2	0	0	0	4	51.76
4	ACGTGGATCCTGAGAACTTCAGG	chr11:5226577	-	50	2	1	0	0	2	31.66
5	GTCTGCCGTTACTGCCCTGTGGG	chr11:5226973	-	60	1	0	0	1	4	46.54
6	CACGTTACCTTGCCCCACAGGG	chr11:5226959	+	60	0	0	1	0	5	61.11
7	TGCTGTTATGGGCAACCCTAAGG	chr11:5226711	-	50	0	1	0	0	5	55.78
8	GGGTTGCCCATAACAGCATCAGG	chr11:5226716	+	55	0	1	0	0	5	54.07
9	CGTTACTGCCCTGTGGGGCAAGG	chr11:5226967	-	65	2	0	0	1	6	46.69
10	TGCCATGAGCCTTCACCTTAGGG	chr11:5226696	+	50	2	1	0	0	5	30.68
11	AGTCTGCCGTTACTGCCCTGTGG	chr11:5226974	-	60	1	0	0	1	7	65.81
12	CCTTGATACCACTGCCAGGG	chr11:5226917	+	55	0	1	0	0	6	64.54
13	GTCCACTCCTGATGCTGTTATGG	chr11:5226723	-	50	0	0	1	1	5	22.29
14	GAAGTTACAGGTGAGTCTATGGG	chr11:5226564	-	45	1	0	0	2	6	50.06
15	CTTAGTGATACTTGTGGGCCAGG	chr11:5225597	+	50	0	0	0	0	9	39.67
16	CACAGACCAGCACGTTGCCAGG	chr11:5225702	+	65	0	0	0	1	8	51.99
17	GGTAGACCACCAGCAGCCTAAGG	chr11:5226783	+	60	2	0	0	1	8	51.85
18	GCAGGCTGCCTATCAGAAAGTGG	chr11:5225639	-	55	0	0	1	0	8	64.86

https://design.synthego.com/#/

Knockout Guide Design

Design top-scoring guide RNAs for gene knockout

Genome

Homo sapiens
Gencode Release 21 (GRCh38)

Popular

Homo SapiensDrosophilaMus Musculus

Gene

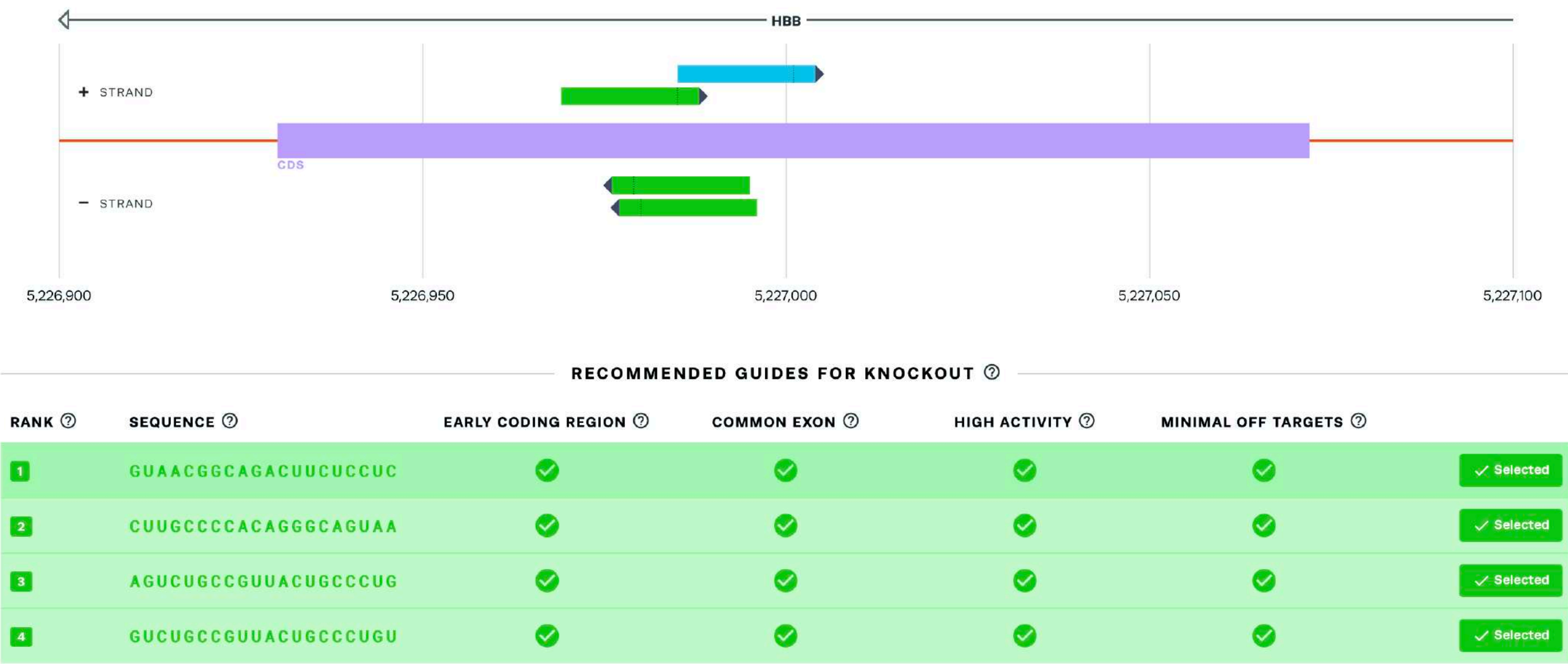
HBB

Nuclease

SpCas9

Search

We found 24 targets and recommend these 4 top-ranked guide RNAs for knocking out the HBB gene in the Homo sapiens genome.



Identify the above guide RNAs for HBB in ChopChop?
What is the rank of the 4 guide RNAs from above on ChopChop?